

When the side of an open tank filled with water is punctured, water flows out according to Bernoulli's equation:

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

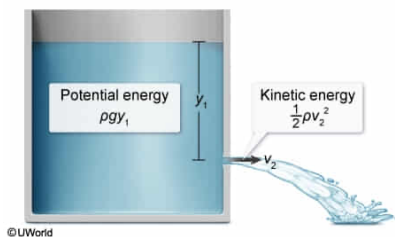
where P is the external pressure, ρ is the density of water (10^3 kg/m^3), v is the flow velocity, g is gravitational acceleration (10 m/s^2), and y is the elevation. If water flows out at a rate of 6 m/s , what is the height of the water above the hole?

- ☐ A. 1.0 m [5%]
☐ B. 1.3 m [16%]
☐ C. 1.6 m [18%]
☒ D. 1.8 m [60%]

Correct

59% answered correctly

Explanation:



Bernoulli's equation is based on the conservation of energy for fluids. The sum of the energy due to the external pressure (P), movement of the fluid ($\frac{1}{2}\rho v^2$), and the weight of the fluid ($\rho g y$) must be equal at any two points. The left side of the equation is set at the surface of the water in the tank (1), and the right side of the equation is set to the hole outside the tank (2):

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2$$

Before solving for the height of the water above the hole, the equation can be simplified by **canceling out terms**. The external pressures P_1 and P_2 are both equal to atmospheric pressure and cancel each other out. The velocity at which the water level falls is much lower than the velocity of the water leaving through the hole ($v_1 \ll v_2$), so v_1 and $\frac{1}{2}\rho v_1^2$ are considered to be negligible and equal to 0. By setting the height of the hole to be the reference elevation, y_2 and $\rho g y_2$ are also equal to 0. Therefore, the equation is greatly simplified:

$$\rho g y_1 = \frac{1}{2}\rho v_2^2$$

This equation shows that the **potential energy above the hole** is equal to the **kinetic energy out the hole**. Canceling out ρ from both sides and using the given values ($g = 10 \text{ m/s}^2$ and $v_2 = 6 \text{ m/s}$), the elevation of the water above the hole is:

$$y_1 = \frac{v_2^2}{2g} = \frac{(6 \text{ m/s})^2}{2(10 \text{ m/s}^2)} = 1.8 \text{ m}$$

Educational objective:

Bernoulli's equation describes the equilibrium of the pressure, kinetic energy, and potential energy between two points in a flowing liquid. When the equation is applied to a punctured tank, the weight of the water above the hole (potential energy) drives the movement of water through the hole (kinetic energy).

Time spent: 0 sec
 Subject:
 Physics

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 System:
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